

Vishal Naveen Akkala

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience developing machine learning models, Generative AI applications, and scalable data solutions across financial services, healthcare, and retail domains. Skilled in predictive modeling, Retrieval-Augmented Generation (RAG), NLP, MLOps, model deployment, and large-scale data processing using Python, SQL, PyTorch, Scikit-Learn, Databricks, AWS, and Azure.

PROFESSIONAL EXPERIENCE

AI/ML Engineer - Capital One

Apr 2025 - Present | FL, USA

- Built Retrieval-Augmented Generation (RAG) applications using LangChain, vector databases, embeddings, and LLMs, improving information discovery across internal knowledge repositories.
- Developed classification, forecasting, and anomaly detection models using Python, Scikit-Learn, XGBoost, and PyTorch, enabling data-driven decision-making.
- Processed 15M+ records monthly through PySpark and Databricks pipelines, accelerating feature engineering across multiple ML initiatives.
- Automated model deployment with MLflow, Docker, GitHub Actions, and cloud services; implemented monitoring for prediction quality, drift, and model health.
- Fine-tuned transformer-based models on domain data to strengthen document classification, information extraction, and search relevance.

ML Engineer - Accenture

Feb 2020 - Dec 2023 | India

- Built end-to-end ML pipelines (ingestion, transformation, feature engineering, training, validation, deployment, monitoring) for financial, healthcare, and retail clients.
- Developed regression, classification, and forecasting models; performed EDA on datasets exceeding 10M records to surface actionable patterns.
- Delivered NLP solutions for sentiment analysis, document categorization, and text classification, reducing manual processing effort.
- Migrated ML workloads to AWS and Azure and built Power BI / Tableau dashboards presenting model outputs and business insights.

EDUCATION

M.S. Information Systems & Operations Management - University of Florida

Dec 2025 | FL, USA

B.Tech. Electrical & Electronics Engineering - National Institute of Technology

Jun 2021 | India

TECHNICAL SKILLS

Languages	Python, SQL, PySpark, R
Machine Learning	Classification, Regression, Forecasting, Anomaly Detection, Recommendation Systems, Feature Engineering, Model Evaluation
Generative AI	LLMs, RAG, LangChain, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Fine-Tuning
NLP	Transformers, Hugging Face, Text Classification, Information Extraction, Sentiment Analysis, NER
Frameworks	Scikit-Learn, PyTorch, TensorFlow, XGBoost, Pandas, NumPy
MLOps	MLflow, Docker, CI/CD, GitHub Actions, Model Monitoring, Experiment Tracking, A/B Testing
Cloud & Data	AWS, Azure, Azure OpenAI, SageMaker, Databricks, Spark, Airflow, ETL
Databases	PostgreSQL, MySQL, SQL Server, MongoDB, Pinecone, ChromaDB
Visualization	Power BI, Tableau